Project name: Automated Login Testing

Module Name: login module

Webpage URI - https://www.google.com,
https://www.saucedemo.com

Testcase ID: TC-15

Testcase: Authentication testing verifies whether valid users can access a system and invalid users are properly rejected.

Code:

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
from selenium.common.exceptions import (
NoSuchElementException,
TimeoutException,
ElementNotInteractableException,
StaleElementReferenceException
)
URL = "https://www.saucedemo.com/"
VALID_USERNAME = "standard_user"
VALID_PASSWORD = "secret_sauce"
def start_browser():
driver = webdriver.Chrome()
driver.maximize_window()
return driver
def login(driver, username, password):
driver.get(URL)
wait = WebDriverWait(driver, 10)
username_box = wait.until(
```

```
EC.visibility_of_element_located(
(By.ID, "user-name")
)
)
password_box = wait.until(
EC.visibility_of_element_located(
(By.ID, "password")
)
)
login_button = wait.until(
EC.element_to_be_clickable(
(By.ID, "login-button")
)
)
username_box.clear()
username_box.send_keys(username)
password_box.clear()
password_box.send_keys(password)
login_button.click()
# TC-01 Valid Login
def test_valid_login():
driver = start_browser()
try:
login(
driver,
VALID_USERNAME,
    VALID_PASSWORD
)
WebDriverWait(driver, 10).until(
    EC.url_contains("inventory")
)
assert "inventory" in driver.current_url
products_title = WebDriverWait(
    driver, 10
).until(
    EC.visibility_of_element_located(
        (By.CLASS_NAME, "title")
```

```
    )
)
assert products_title.text == "Products"
print("TC-01 Valid Login: PASS")
print("Authentication Assertion: PASS")
except AssertionError:
print("TC-01 Valid Login: FAIL")
except NoSuchElementException:
print("NoSuchElementException handled")
except TimeoutException:
print("TimeoutException handled")
except ElementNotInteractableException:
print(
    "ElementNotInteractableException handled"
)
except StaleElementReferenceException:
print(
    "StaleElementReferenceException handled"
)
finally:
driver.quit()


# TC-02 Invalid Password
def test_invalid_password():
    driver = start_browser()


try:
    login(
        driver,
        VALID_USERNAME,
        "wrong_password"
    )
error_message = WebDriverWait(
    driver, 10
).until(
```

```
    EC.visibility_of_element_located(
        (
            By.CSS_SELECTOR,
            "h3[data-test='error']"
        )
    )
)
assert (
        "Username and password do not match"
        in error_message.text
)
print("TC-02 Invalid Password: PASS")

except AssertionError:
print("TC-02 Invalid Password: FAIL")

except TimeoutException:
print("Error message was not displayed")

finally:
driver.quit()


# TC-03 Invalid Username
def test_invalid_username():
    driver = start_browser()

    try:
        login(
            driver,
            "wrong_user",
            VALID_PASSWORD
        )

        error_message = WebDriverWait(
            driver, 10
        ).until(
```

```
            EC.visibility_of_element_located(
                (
                    By.CSS_SELECTOR,
                    "h3[data-test='error']"
                )
            )
        )

        assert (
                "Username and password do not match"
                in error_message.text
        )


print("TC-03 Invalid Username: PASS")
except AssertionError:
print("TC-03 Invalid Username: FAIL")
except TimeoutException:
print("Error message was not displayed")
finally:
driver.quit()


# TC-04 Empty Credentials
def test_empty_credentials():
    driver = start_browser()


try:
    driver.get(URL)
login_button = WebDriverWait(
    driver, 10
).until(
    EC.element_to_be_clickable(
        (By.ID, "login-button")
    )
)
```

```
login_button.click()
error_message = WebDriverWait(
    driver, 10
).until(
    EC.visibility_of_element_located(
        (
            By.CSS_SELECTOR,
            "h3[data-test='error']"
        )
    )
)
assert (
        "Username is required"
        in error_message.text
)
print("TC-04 Empty Credentials: PASS")
except AssertionError:
print("TC-04 Empty Credentials: FAIL")
except TimeoutException:
print("Validation message was not displayed")
finally:
driver.quit()

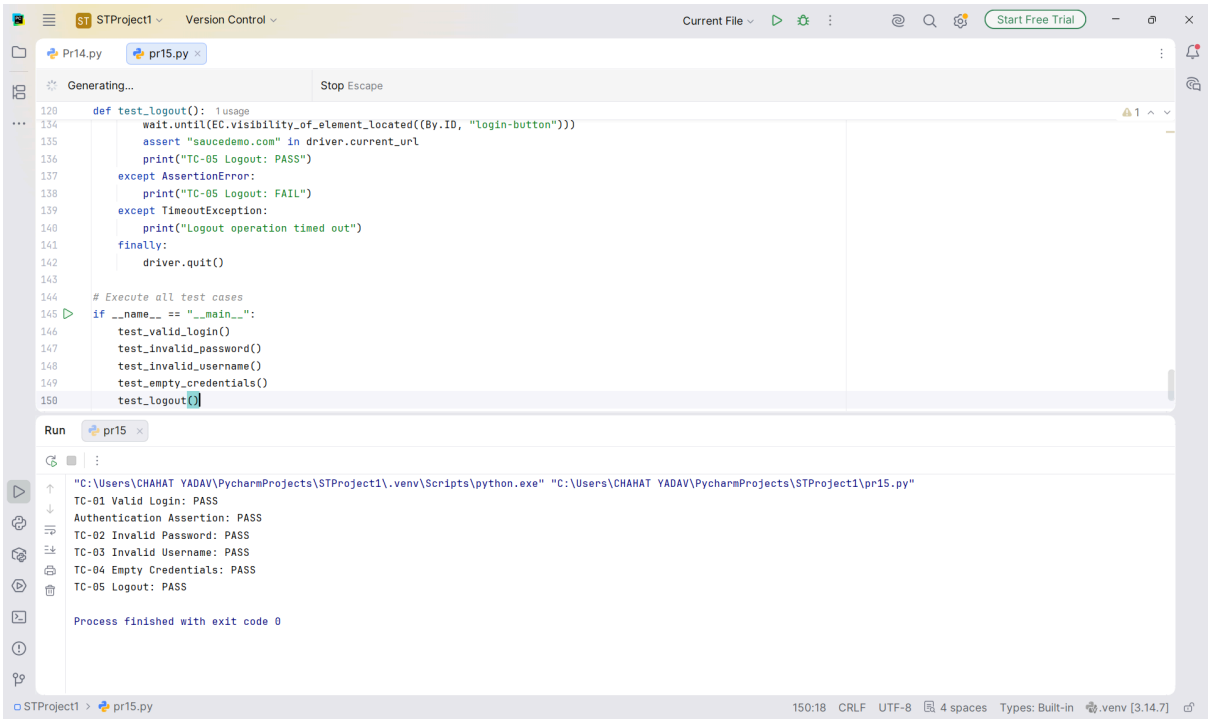

# TC-05 Logout
def test_logout():
    driver = start_browser()

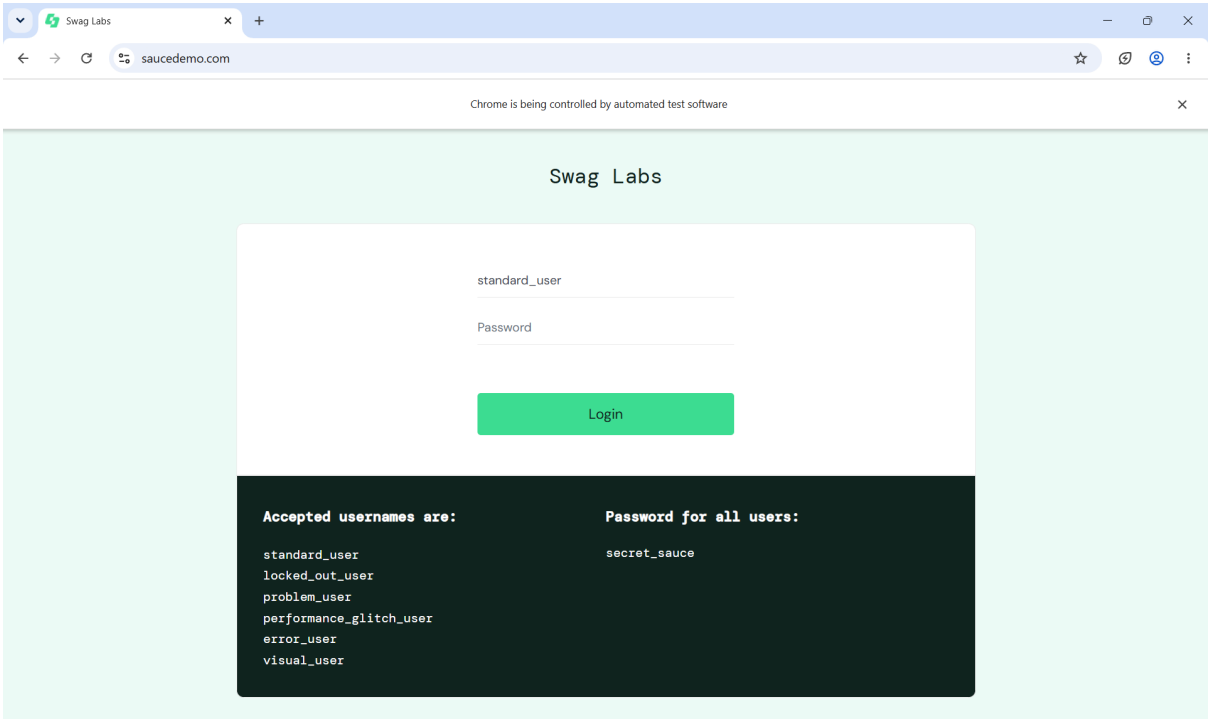

try:
    login(
        driver,
        VALID_USERNAME,
        VALID_PASSWORD
    )
WebDriverWait(driver, 10).until(
    EC.url_contains("inventory")
```

```
)
menu_button = WebDriverWait(
    driver, 10
).until(
    EC.element_to_be_clickable(
        (By.ID, "react-burger-menu-btn")
    )
)
menu_button.click()
logout_button = WebDriverWait(
    driver, 10
).until(
    EC.element_to_be_clickable(
        (By.ID, "logout_sidebar_link")
    )
)
logout_button.click()
WebDriverWait(driver, 10).until(
    EC.visibility_of_element_located(
        (By.ID, "login-button")
    )
)
assert "saucedemo.com" in driver.current_url
print("TC-05 Logout: PASS")
except AssertionError:
print("TC-05 Logout: FAIL")
except TimeoutException:
print("Logout operation timed out")
finally:
driver.quit()
# Execute all test cases
test_valid_login()
test_invalid_password()
test_invalid_username()
test_empty_credentials()
test_logout()
```

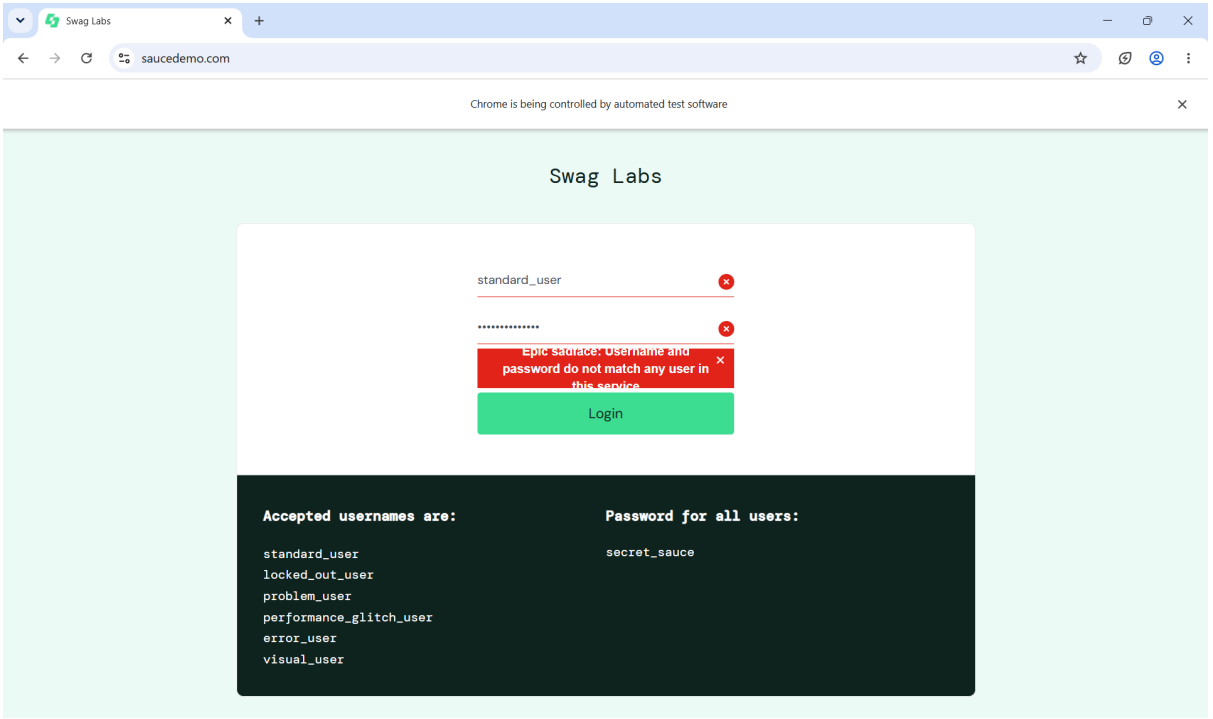
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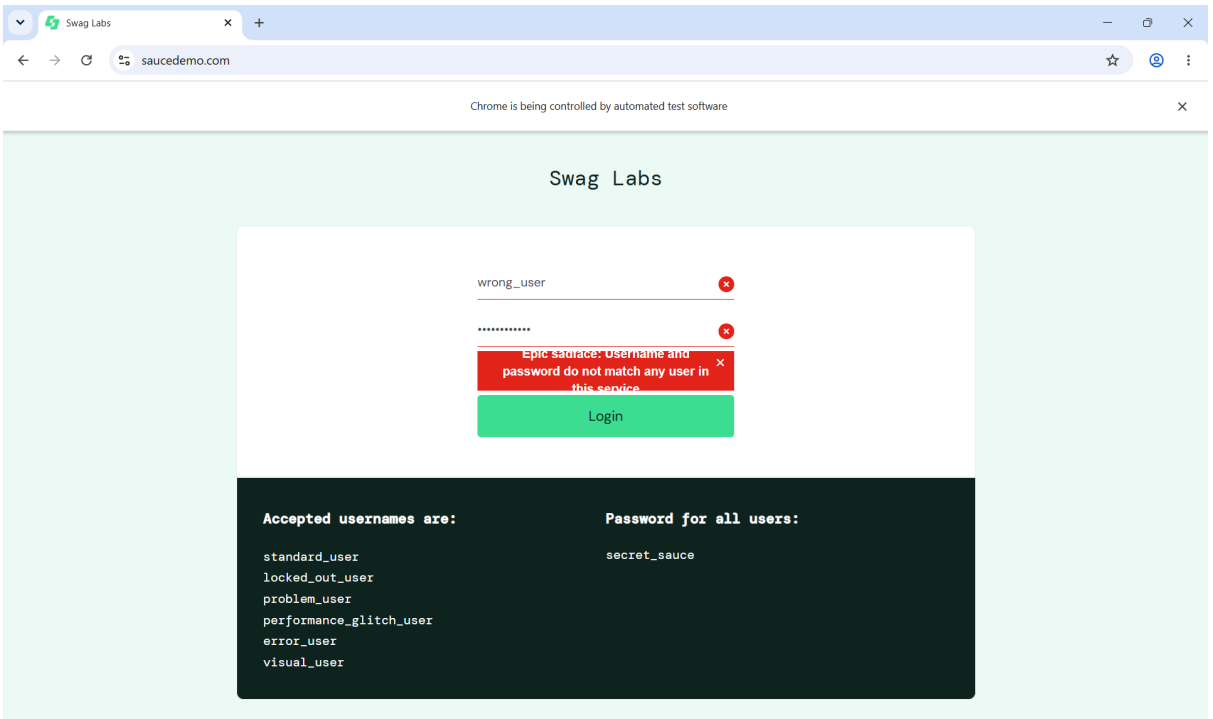
Screenshots:



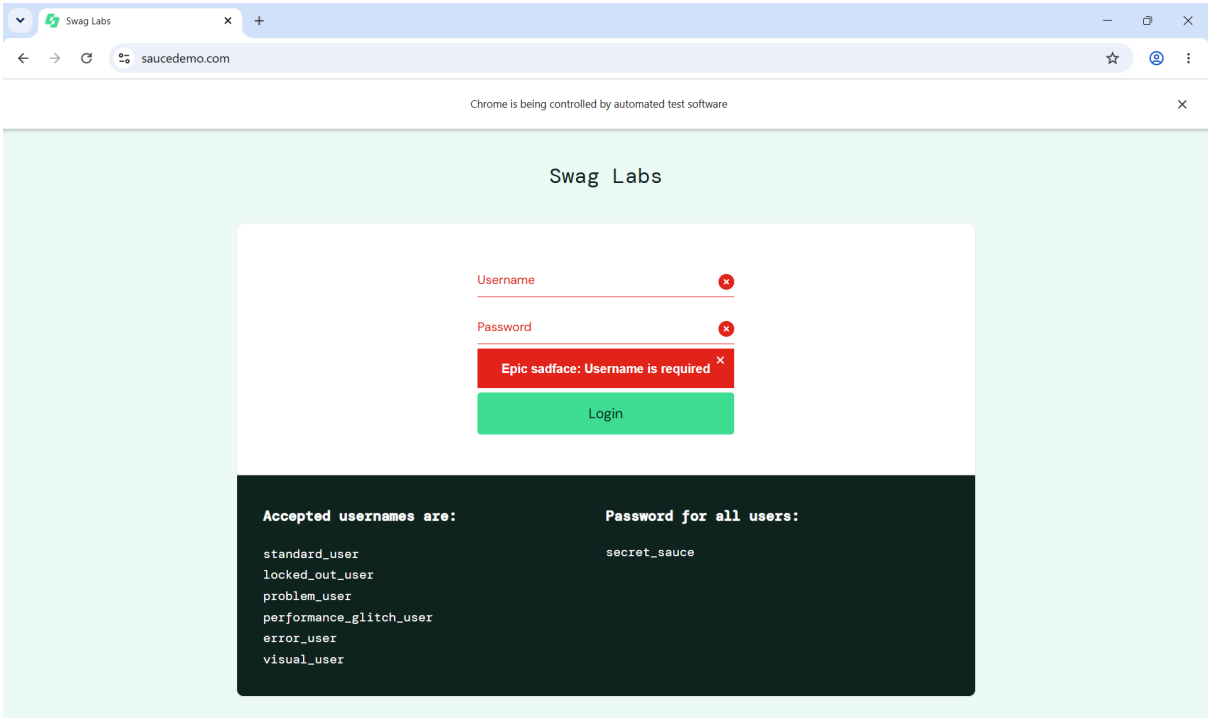


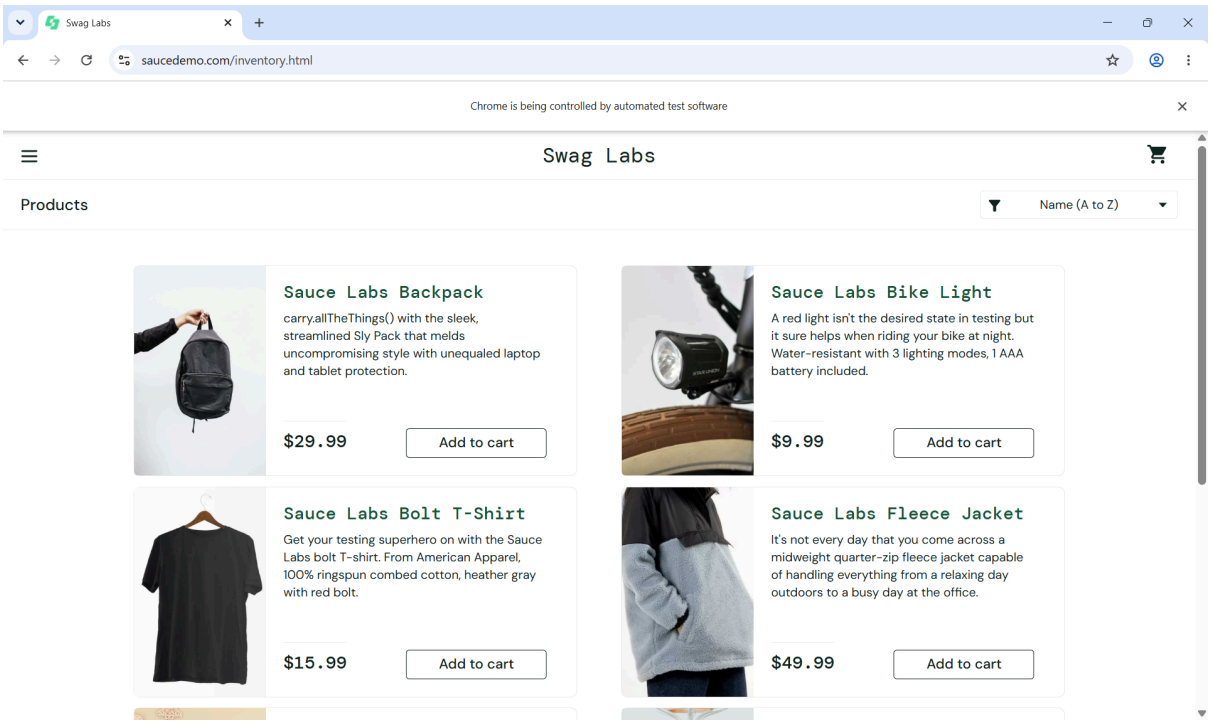
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Swag Labs
saucedemo.com
Chrome is being controlled by automated test software
Swag Labs
standard_user
Epic sadface: Username and password do not match any user in this service
Login
Accepted usernames are:
standard_user
locked_out_user
problem_user
performance_glitch_user
error_user
visual_user
Password for all users:
secret_sauce

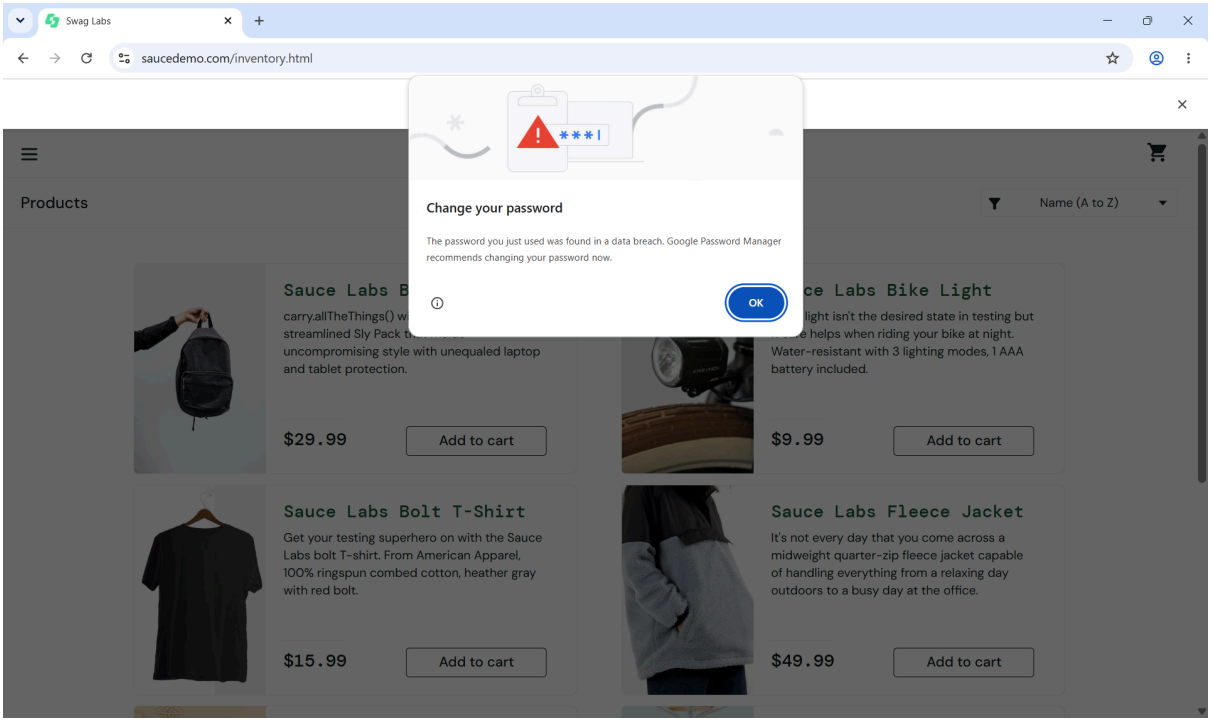

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Password for all users:
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Swag Labs
saucedemo.com
Chrome is being controlled by automated test software
Swag Labs
Username
Password
Epic sadface: Username is required
Login
Accepted usernames are:
standard_user
locked_out_user
problem_user
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visual_user
Password for all users:
secret_sauce


Swag Labs
saucedemo.com/inventory.html
Chrome is being controlled by automated test software
Swag Labs
Products
Name (A to Z)
Sauce Labs Backpack
carry.allTheThings() with the sleek, streamlined Sly Pack that melds uncompromising style with unequaled laptop and tablet protection.
$29.99
Add to cart
Sauce Labs Bike Light
A red light isn't the desired state in testing but it sure helps when riding your bike at night. Water-resistant with 3 lighting modes, 1 AAA battery included.
$9.99
Add to cart
Sauce Labs Bolt T-Shirt
Get your testing superhero on with the Sauce Labs bolt T-shirt. From American Apparel, 100% ringspun combed cotton, heather gray with red bolt.
$15.99
Add to cart
Sauce Labs Fleece Jacket
It's not every day that you come across a midweight quarter-zip fleece jacket capable of handling everything from a relaxing day outdoors to a busy day at the office.
$49.99
Add to cart

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Swag Labs
saucedemo.com/inventory.html
Change your password
The password you just used was found in a data breach. Google Password Manager recommends changing your password now.
OK
Products
Name (A to Z)
Sauce Labs Bike Light
Water-resistant with 3 lighting modes, 1 AAA battery included.
$9.99
Add to cart
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Get your testing superhero on with the Sauce Labs bolt T-shirt. From American Apparel, 100% ringspun combed cotton, heather gray with red bolt.
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